Jinwoo Jeon

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Technical Skills

Programming: C++, Python, MATLAB, CUDA
Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, Docker, Git
Research Areas: SLAM, 3D LiDAR, Point Cloud Registration, Semantic Segmentation, Sensor Fusion, Deep Learning

Selected Projects

VINS-Fusion-ROS2 [★ **378**] | – Developed and maintained the first open-source ROS2 port of VINS-Fusion, supporting ROS2 execution and optional GPU-enabled feature tracking.
KAIST VIO dataset [★ **110**] *RA-L 2021* | – UAV-based visual-inertial odometry benchmark dataset for evaluating VO/VIO robustness under challenging motions, including fast motion, rotational motion, and pure rotation.

Publications

1. **Jinwoo Jeon**, Dong-Uk Seo, Hyun Myung, "DUET-SLAM: Dual-Flow Consistency for Dense Monocular SLAM in Dynamic Environments with Geometry-aware Foundation Model," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, Accepted. To appear.
2. Dong-Uk Seo, Jiwon Park, Jei Kong, **Jinwoo Jeon**, Hyun Myung, "Leveraging Planar Regularity for Gaussian Splatting SLAM," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, Accepted. To appear.
3. Junwan Choi, Woongrae Jo, Dong-Uk Seo, **Jinwoo Jeon**, Hyun Myung, "Robust Structureless Monocular Visual-Inertial Initialization Exploiting Line Features and Vanishing Points," *IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Accepted. To appear.
4. Dong-Uk Seo, **Jinwoo Jeon**, Eungchang Mason Lee, Hyun Myung, "GaussianFlow SLAM: Monocular Gaussian Splatting SLAM Guided by GaussianFlow," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 11, no. 6, pp. 7644-7651, 2026.
5. **Jinwoo Jeon**, Dong-Uk Seo, Eungchang Mason Lee, Hyun Myung, "AIM-SLAM: Dense Monocular SLAM via Adaptive and Informative Multi-View Keyframe Prioritization with Foundation Model," *IEEE Int. Conf. Robot. Automat. (ICRA)*, 2026. Accepted. To appear.
6. K. C. Marsim, **Jinwoo Jeon**, Yeeun Kim, Myungwoo Jeong, Hyun Myung, "CHADET: Cross-Hierarchical-Attention for Depth-Completion Using Unsupervised Lightweight Transformer," *IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, 2025, pp. 4931-4938.
7. **Jinwoo Jeon**, Myungwoo Jeong, Hyun Myung, "Fuzzy Logic and Neural Network-based Intelligent Control System for Quadruped Robot on Extreme Terrain," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2023, pp. 1885-1889.
8. Byeongho Yu, Yeeun Kim, **Jinwoo Jeon**, I Made Aswin Nahrendra, and Hyun Myung, "A Robust State Estimator for Legged Robots using Visual-Inertial-Leg Kinematics in the Wild," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2022.
9. Eungchang Mason Lee, **Jinwoo Jeon**, Hyun Myung, "TROT-Q: Traversability and Obstacle Aware Target Tracking System for Quadruped Robots," *ASCC*, 2022, pp. 2480-2484.
10. **Jinwoo Jeon**, Hyunjun Lim, Dong-Uk Seo, Hyun Myung, "Struct-MDC: Mesh-Refined Unsupervised Depth Completion Leveraging Structural Regularities from Visual SLAM," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 3, pp. 6391-6398, 2022.
11. Hyunjun Lim, **Jinwoo Jeon**, Hyun Myung, "UV-SLAM: Unconstrained Line-based SLAM Using Vanishing Points for Structural Mapping," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 2, pp. 1518-1525, 2022.
12. Seunghyun Lee, Sumin Chang, Eungchang Mason Lee, Junho Choi, **Jinwoo Jeon**, Seoktae Kim, Hyun Myung, "TPL: Trajectory Planner for Target Tracking in Low-light Environments," *Proc. Int. Conf. Robot Intell. Tech. Appl. (RiTA)*, 2021, pp. 12-23.
13. Eungchang Mason Lee, Dong-Uk Seo, **Jinwoo Jeon**, Hyun Myung, "QR-SCAN: Traversable Region Scan for Quadruped Robot Exploration Using Lightweight Precomputed Trajectory," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2021, pp. 957-961.
14. ChanYoung Kim, EungChang Mason Lee, JunHo Cho, **Jinwoo Jeon**, Seoktae Kim, Hyun Myung, "ROLAND: Robust Landing of UAV on Moving Platform Using Object Detection and UWB-based Extended Kalman Filter," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2021, pp. 249-254.
15. **Jinwoo Jeon***, Sungwook Jung*, Eungchang Lee, Duckyu Choi, Hyun Myung, "Run Your Visual-Inertial Odometry on NVIDIA Jetson: Benchmark Tests on a Micro Aerial Vehicle," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 6, no. 3, pp. 5332-5339, 2021.

Honors & Awards

2026 – 1st Prize, SLAM category of Hilti × Trimble Challenge'26 in IEEE ICRA among 62 international teams
2026 – Travel Grant for IEEE ICRA'26, Institute of Control, Robotics and Systems (ICROS)

Education

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
Ph.D. in Electrical Engineering *Feb. 2023 – Present*

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
M.S. in Electrical Engineering *Mar. 2021 – Feb. 2023*

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
B.S. in Electrical Engineering (Minor in Computer Science) *Mar. 2016 – Feb. 2021*